

ORIGINAL CONTRIBUTION

Ticagrelor-Aspirin With Thrombolysis by Infarct Patterns in Moderate Ischemic Stroke: TAPIS Subgroup Analysis

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BACKGROUND: Whether diffusion-weighted imaging infarction patterns influence the efficacy and safety of early oral ticagrelor-aspirin with intravenous thrombolysis in moderate acute ischemic stroke is uncertain. We assessed treatment effects by infarction pattern in this thrombolysis setting.

METHODS: This was a subgroup analysis of the TAPIS trial (Ticagrelor With Aspirin Dual Antiplatelet Therapy Combined With Intravenous Thrombolysis in Patients With Ischemic Stroke), a 60-center, randomized, double-blind trial in China. Patients aged 18 to 80 years with noncardioembolic stroke (National Institutes of Health Stroke Scale score 4–10) who received intravenous thrombolysis within 4.5 hours of onset were randomized within 6 hours to ticagrelor-aspirin or placebo. Diffusion-weighted imaging patterns were centrally adjudicated as multiple acute infarcts (MAIs; ≥ 2 lesions) or non-MAIs (single infarction or diffusion-weighted imaging-negative). The primary efficacy outcome was modified Rankin Scale score of 0 to 1 at 90 days; safety outcome was symptomatic intracranial hemorrhage. Treatment-by-pattern interaction was assessed.

RESULTS: Among 1307 patients (94.6% of full analysis set; median age, 65.5 years; 71.2% male; median National Institutes of Health Stroke Scale score, 6.0), 409 (31.3%) had MAIs and 898 (68.7%) non-MAIs (20.8% diffusion-weighted imaging-negative, 79.2% single infarction). MAIs were associated with poorer prognosis than non-MAIs (excellent outcome: 58.9% versus 69.2%; adjusted risk ratio, 0.92 [95% CI, 0.84–1.00]). Ticagrelor-aspirin was associated with a higher likelihood of excellent outcome versus placebo in MAIs (64.4% versus 53.4%; risk ratio, 1.21 [95% CI, 1.02–1.42]), whereas no significant benefit was observed in non-MAIs (71.2% versus 67.1%; risk ratio, 1.06 [95% CI, 0.97–1.16]; $P_{\text{interaction}}=0.18$). Early neurological improvement showed significant heterogeneity by infarction pattern ($P_{\text{interaction}}=0.02$); the treatment-by-pattern interaction for the primary outcome was nominally significant after adjusting for rescue tirofiban use ($P=0.04$). Symptomatic intracranial hemorrhage occurred in MAIs: 1 of 205 (0.5%) versus 2 of 204 (1.0%); non-MAIs: 1 of 448 (0.2%) versus 1 of 450 (0.2%).

CONCLUSIONS: In thrombolized patients with moderate noncardioembolic stroke, MAIs were associated with poorer functional outcomes. Early ticagrelor-aspirin showed directionally consistent benefits without excess bleeding in patients with MAIs, though the primary interaction was not significant. These hypothesis-generating findings may inform future trials of diffusion-weighted imaging infarction pattern as an imaging biomarker to guide antiplatelet therapy in early reperfusion.

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GRAPHIC ABSTRACT: A graphic abstract is available for this article.

Key Words: aspirin ■ infarction ■ intracranial hemorrhage ■ ischemic stroke ■ ticagrelor

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Nonstandard Abbreviations and Acronyms

DWI	diffusion-weighted imaging
ECASS III	European Cooperative Acute Stroke Study III
GUSTO	Global Utilization of Streptokinase and Tissue Plasminogen Activator for Occluded Coronary Arteries
IQR	interquartile range
IVT	intravenous thrombolysis
MAI	multiple acute infarcts
mRS	modified Rankin Scale
NIHSS	National Institutes of Health Stroke Scale
PAI	perforator artery infarction
RR	risk ratio
slCH	symptomatic intracranial hemorrhage
SITS-MOST	Safe Implementation of Thrombolysis in Stroke-Monitoring Study
TAPIS	Ticagrelor With Aspirin Dual Antiplatelet Therapy Combined With Intravenous Thrombolysis in Patients With Ischemic Stroke
TOAST	Trial of Org 10172 in Acute Stroke Treatment

Intravenous thrombolysis (IVT) is the standard reperfusion therapy for eligible patients with acute ischemic stroke.^{1–3} However, incomplete recanalization, early reocclusion, or persistent microvascular thrombosis occur in approximately one-third of patients after IVT and may contribute to neurological deterioration and poor functional outcomes.^{4,5} This prothrombotic state arises partly from platelet activation and the release of procoagulant factors induced by the fibrinolytic process itself.^{6–8} Therefore, early antiplatelet therapy after IVT represents a biologically plausible strategy to reduce platelet-rich thrombus formation and maintain downstream microvascular patency.⁹ Identifying patients most likely to benefit from this strategy remains an important clinical challenge.

Diffusion-weighted imaging infarction patterns reflect important heterogeneity in stroke mechanisms and thromboembolic activity.^{10–12} Multiple acute infarcts (MAIs) are commonly associated with unstable embolic sources, including large artery atherosclerosis and cardioembolism, and have been associated with an increased risk of recurrent ischemic stroke and poorer clinical outcomes in previous studies, potentially reflecting ongoing thromboembolic activity and a higher burden of vascular instability.^{11,13} Compared with single acute infarction or diffusion-weighted imaging (DWI)–negative phenotypes, MAIs may represent a state of heightened platelet-mediated thromboembolic burden.^{13–16}

Prior studies in patients with minor stroke or transient ischemic attack have suggested imaging-defined phenotypes may be associated with differential responses to dual antiplatelet therapy. In the CHANCE imaging substudy, clopidogrel-aspirin reduced stroke recurrence more substantially in patients with MAI (hazard ratio, 0.50; $P_{\text{interaction}}=0.04$).¹⁴ Similarly, the post hoc analysis of the POINT trial showed the benefit of clopidogrel-aspirin for reducing ischemic stroke was confined to patients with an infarct on index imaging.¹⁷ The CHANCE-2 trial enrolled CYP2C19 loss-of-function carriers. In 1628 DWI-positive patients, ticagrelor-aspirin was superior to clopidogrel-aspirin.¹⁸ Together, these findings suggest imaging phenotypes may help identify patients with greater platelet-mediated ischemic vulnerability.

However, existing evidence is derived almost entirely from nonthrombolized patients with mild neurological deficits. The postthrombolysis setting involves complex biological and clinical considerations, mainly including reperfusion injury, risk of hemorrhagic transformation, and thrombolysis-induced platelet activation.^{19,20} Whether index infarction patterns influence the efficacy and safety of intensive oral antiplatelet therapy combined with IVT therefore remains uncertain. The TAPIS trial (Ticagrelor With Aspirin Dual Antiplatelet Therapy Combined With Intravenous Thrombolysis in Patients With Ischemic Stroke) recently showed that early oral dual antiplatelet therapy (within 6 hours of onset) improved the likelihood of excellent functional outcome at 90 days in patients treated with IVT for moderate non-cardioembolic ischemic stroke, with no significant between-group difference in symptomatic intracranial hemorrhage (slCH), despite wide CIs.²¹ In this prespecified imaging subgroup analysis of TAPIS, we evaluated the prognostic significance of DWI infarction patterns and whether the efficacy and safety of ticagrelor-aspirin varied by index infarction pattern in the thrombolysis setting. We hypothesized that patients with MAIs, given their greater thromboembolic burden, might derive greater benefit from early oral potent P2Y₁₂ inhibition.

METHODS

Data Availability

Individual participant data are not publicly available owing to patient privacy and trial governance restrictions. Deidentified analytic data sets may be accessible to qualified researchers on reasonable request to the corresponding author, subject to a formal proposal approved by the trial steering committee.

STUDY DESIGN, PARTICIPANTS, AND PROCEDURES

The TAPIS trial was an investigator-initiated, multicenter, randomized, double-blind, placebo-controlled trial conducted at 60 sites in China.²¹ The trial was approved

by the ethics committee of Beijing Tiantan Hospital (KY2023-200-03) and all participating centers. Written informed consent was obtained from all participants or their legally authorized representatives before enrollment. Eligible participants were aged 18 to 80 years, had a prethrombolysis National Institutes of Health Stroke Scale (NIHSS) score of 4 to 10 with at least 1 point on a limb motor item, and had received or were intended to receive IVT with alteplase or tenecteplase within 4.5 hours after symptom onset. Key exclusion criteria included planned endovascular therapy; prestroke modified Rankin Scale score ≥ 2 ; antiplatelet therapy after symptom onset before randomization; preexisting atrial fibrillation or anticoagulant use; conditions associated with elevated bleeding risk, including coagulation disorders, systemic bleeding, thrombocytopenia, neutropenia, hepatic or renal insufficiency, and intracranial hemorrhagic lesions; and deemed unsuitable by the treating investigator. A complete list of eligibility criteria is provided in the Appendix in the [Supplemental Material](#). The randomization sequence was generated centrally by an independent statistician using a site-stratified, blocked method with a fixed block size of 4, with allocation concealment maintained and the sequence concealed from enrolling personnel until assignment. Patients were randomized 1:1, and study medication was administered orally at randomization, which could occur before, during, or after IVT. The ticagrelor-aspirin group received ticagrelor 180 mg plus aspirin 100 mg on day 1, followed by ticagrelor 90 mg twice daily plus open-label aspirin 100 mg daily on days 2 to 7; the placebo group received matching placebo on day 1 and ticagrelor placebo plus open-label aspirin on days 2 to 7; active and placebo tablets were identical in appearance, taste, and packaging to maintain blinding. All participants received open-label aspirin 100 mg daily from days 8 to 90. Rescue antiplatelet therapy with tirofiban was permitted for protocol-defined neurological deterioration within 7 days (increase in NIHSS score ≥ 4 from nadir or return to pre-thrombolysis level) after intracranial hemorrhage had been excluded by imaging; other off-protocol antiplatelet agents were prohibited. An independent clinical event adjudication committee and data and safety monitoring board oversaw trial conduct and safety assessment. All investigators and participants were blinded to treatment assignment; unblinding was permitted only with Steering Committee approval for serious adverse events requiring treatment disclosure. Further details of the TAPIS trial design have been published previously.²²

Data Collection and Imaging Analysis

Magnetic resonance imaging scans with DWI sequences ($b \geq 1000$ s/mm²) were collected from participating centers in Digital Imaging and Communications in Medicine format and centrally analyzed at the Tiantan

Neuroimaging Center of Excellence, Beijing Tiantan Hospital. Infarction patterns were classified according to lesion number and topographical distribution. MAIs were defined as ≥ 2 anatomically distinct acute lesions separated spatially or appearing discrete on contiguous DWI slices. Non-MAIs included single acute infarction and DWI-negative stroke. Single acute infarction was defined as a lesion confined to a single contiguous arterial territory without interruption on adjacent DWI slices (slice thickness ≤ 5 –6 mm). According to anatomic location, single acute infarction was further categorized as perforator artery infarction (PAI) or non-PAI. PAI referred to a subcortical infarction within the perforator territories of the middle cerebral, basilar, or posterior cerebral artery, encompassing the basal ganglia, internal capsule, thalamus, or ventral pons. PAI lesions were further subclassified by lesion size and morphology. Single small subcortical infarction denoted a subcortical lesion with a maximum diameter ≤ 20 mm. Following the Caplan definition, branch atheromatous disease indicated a larger subcortical lesion with a comma-shaped appearance, fan-shaped coronal extension, or involvement of 3 or more consecutive axial slices, in the absence of significant parent artery stenosis ($\leq 50\%$). DWI-negative stroke was defined as a clinical diagnosis of acute ischemic stroke without a definite hyperintense lesion on acute DWI and with isointense or hyperintense signal on the apparent diffusion coefficient map (Figure 1). All clinical and imaging data were independently reviewed by 2 experienced neurologists (X.Z. and Q.W.) who were blinded to index clinical characteristics and outcome information. Disagreements were resolved by adjudication from an expert panel. Inter-rater agreement for infarction pattern classification was good (Cohen $\kappa = 0.88$ [95% CI, 0.83–0.91]). Patients were not involved in the design, conduct, or reporting of this imaging substudy.

Outcomes and Measures

The primary efficacy outcome was excellent functional outcome at 90 days, defined as modified Rankin Scale (mRS) score 0 to 1. Secondary efficacy outcomes included functional independence (mRS score 0–2) at 90 days; ordinal mRS score distribution to assess shift toward lower disability; early neurological improvement (≥ 4 -point decrease in NIHSS score from baseline at day 7); quality of life at 90 days using the EQ-5D-5L and EQ-5D VAS; activities of daily living independence (Barthel Index 95–100); and recurrent ischemic stroke within 90 days.

The primary safety outcome was sICH at 36 hours, defined by ECASS III (European Cooperative Acute Stroke Study III) criteria as any intracranial hemorrhage associated with an NIHSS score increase of ≥ 4 points or death where hemorrhage was the predominant cause of neurological deterioration. Secondary safety outcomes

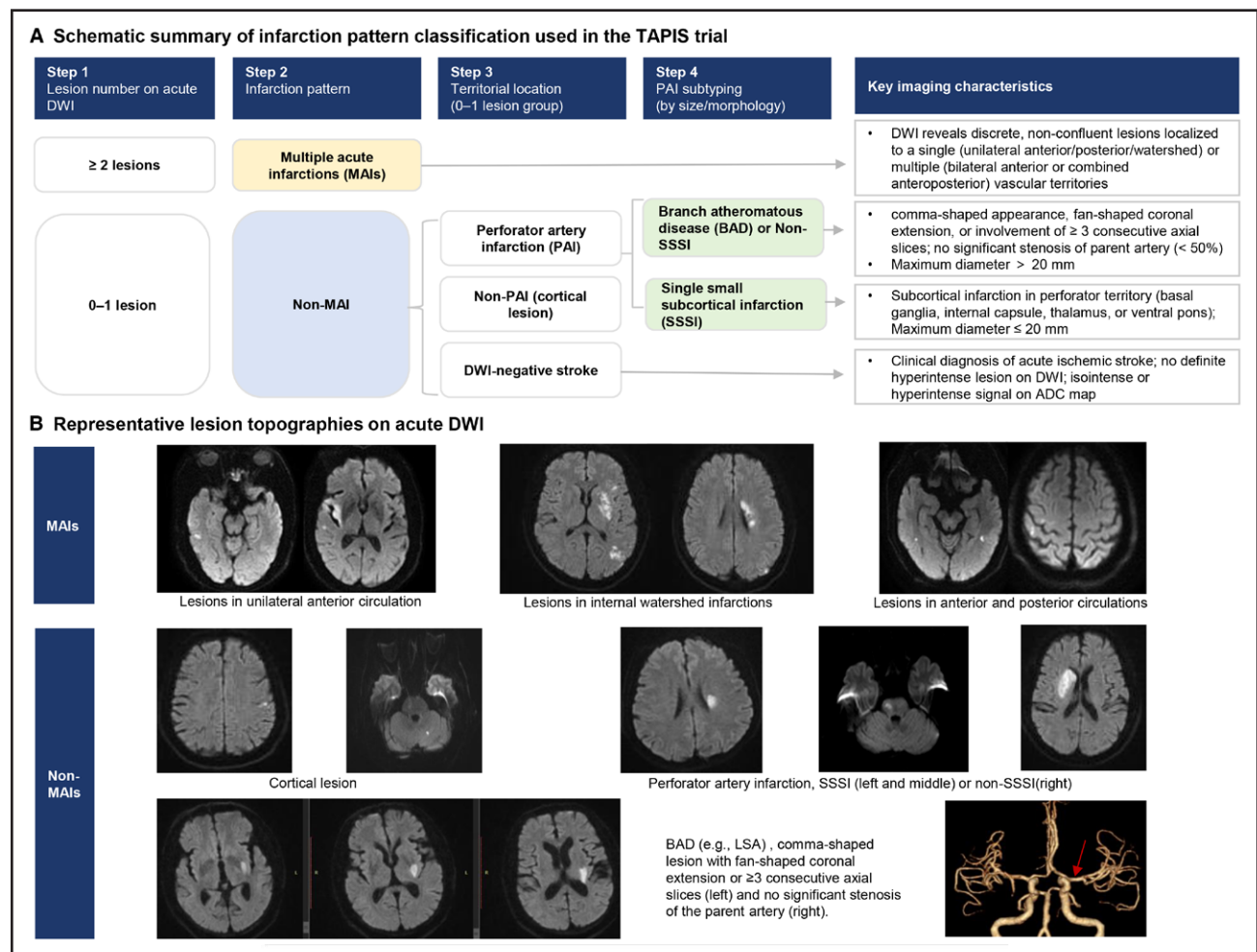


Figure 1. Schematic summary of infarction pattern classification and representative lesion topographies in the TAPIS trial (Ticagrelor With Aspirin Dual Antiplatelet Therapy Combined With Intravenous Thrombolysis in Patients With Ischemic Stroke).

A, Diffusion-weighted imaging (DWI) infarction patterns were classified as multiple acute infarcts (MAIs) or non-MAIs. MAI was defined as 2 or more anatomically distinct acute lesions on DWI. Non-MAI included single acute infarction (SAI) or DWI-negative stroke. SAI was further categorized by anatomic location as perforator artery infarction (PAI) or non-PAI; PAI was subclassified as single small subcortical infarction (SSSI) or branch atheromatous disease (BAD) based on lesion size and morphology. DWI-negative stroke was defined as a clinical diagnosis of acute ischemic stroke without a corresponding hyperintense signal on DWI. **B**, Representative DWI images illustrate the classification: MAI lesions, such as unilateral anterior circulation, internal watershed, or combined anterior and posterior circulation lesions; non-MAI lesions, such as solitary cortical infarcts, isolated perforator artery lesions, and BAD.

included sICH at 7 days; sICH by SITS-MOST (Safe Implementation of Thrombolysis in Stroke-Monitoring Study) criteria (including parenchymal hematoma type 2) at 36 hours and 7 days; any bleeding within 90 days by GUSTO (Global Utilization of Streptokinase and Tissue Plasminogen Activator for Occluded Coronary Arteries) criteria; and investigator-reported adverse events within 90 days assessed by CTCAE version 5.0.

Statistical Analysis

The parent trial was designed with 1380 patients to provide ≈90% power to detect a 9% absolute difference in the primary outcome, assuming a 60% rate in the control group and a 10% dropout rate. A planned interim analysis of the primary efficacy outcome was canceled

by the DSMB, which at that time reviewed only the overall incidences of safety end points and, given that 1238 participants had already been enrolled, recommended continuing the trial to full recruitment.²¹

Primary efficacy analyses followed the intention-to-treat principle and included all randomized patients with available infarction pattern data and at least 1 90-day mRS score assessment. Missing primary outcome data were imputed using last observation carried forward (60-day, 30-day, or 7-day mRS score in descending order). For deceased patients, NIHSS was set to 42, EQ-5D-5L to −0.391, EQ-5D VAS to 0, and Barthel Index to 0. Categorical variables were presented as frequencies and percentages and were compared using the χ^2 test or Fisher exact test, as appropriate. Continuous variables were presented as median with interquartile range and

compared using the Wilcoxon rank-sum test. We used a mixed-effects model with a random intercept for pooled study centers to account for potential site-level clustering across centers. Before the unmasking of treatment groups, sites with <20 participants were pooled within the same geographic region, ensuring that the pooled sites were of adequate size for statistical analysis. Binary efficacy outcomes, including mRS score 0 to 1 at 90 days, mRS score 0 to 2 at 90 days, Barthel Index ≥ 95 at 90 days, and early neurological improvement at 7 days, were analyzed using generalized linear models with a log link and binomial error distribution to estimate risk ratios (RR) with 95% CIs. Ordinal distribution of mRS scores at 90 days was analyzed using proportional odds models after verification of the proportional odds assumption, and treatment effects were reported as common odds ratios. Continuous outcomes (EQ-5D index score and EQ-5D VAS) were compared using the Hodges-Lehmann method to estimate median differences. Time to recurrent ischemic stroke within 90 days was analyzed using Cox proportional hazards models, and the proportional hazards assumption was verified using Schoenfeld residuals. Effect modification by infarction pattern was assessed by including a treatment-by-infarction pattern interaction term in the models; corresponding *P* values for the interaction are reported. Prognostic models were adjusted for age, sex, baseline NIHSS score, and TOAST (Trial of Org 10172 in Acute Stroke Treatment) classification, in addition to baseline variables with univariable $P \leq 0.10$.

We performed sensitivity analyses in the per-protocol population. To address the potential confounding effect of rescue antiplatelet therapy, we conducted a post-hoc sensitivity analysis applying a composite variable strategy in accordance with the ICH E9 (R1) addendum. For the binary efficacy end point, patients who received rescue therapy were classified as having a poor outcome (mRS score 2–6). For the ordinal analysis, these patients were assigned an mRS score of 5 at subsequent visits, or 6 in the event of death. We also examined treatment effects stratified by the timing of randomization relative to thrombolysis and across constituent infarction patterns within the non-MAI subgroup. All analyses beyond the primary efficacy end point were exploratory and not adjusted for multiplicity; therefore, *P* values should be interpreted as nominal and the findings as hypothesis-generating. All tests were 2-sided, with $P < 0.05$ considered statistically significant. All analyses were performed using SAS software, version 9.4 (SAS Institute, Cary, NC).

RESULTS

Baseline Characteristics

Between April 3, 2024, and September 30, 2025, the TAPIS trial enrolled 1382 patients with acute ischemic

stroke. After excluding 75 patients without baseline DWI data, 1307 (94.6%) of the full analysis set were included (Figure S1); baseline characteristics by imaging availability are in Table S1. The median onset-to-scan interval was 14.6 hours (interquartile range [IQR], 3.2–28.7), with corresponding IVT-to-scan and randomization-to-scan intervals of 12.2 hours (IQR, 0.1–26.0) and 11.2 hours (IQR, –0.9 to 24.9), respectively. Patients without DWI lesion detection underwent imaging slightly later than those with visible lesions: the median IVT-to-scan interval was 17.6 hours (IQR, 0.4–26.6) versus 10.6 hours (IQR, 0.1–25.9; $P=0.05$), and the median randomization-to-scan interval was 16.0 hours (IQR, –0.5 to 26.2) versus 10.0 hours (IQR, –1.0 to 24.8; $P=0.03$).

Among the included patients, median age was 65.5 years (IQR, 58.2–72.0), 71.2% were male, and median NIHSS score was 6.0 (IQR, 6.0–7.0). Overall, 898 patients (68.7%) did not have MAIs and 409 (31.3%) had MAIs. Within the non-MAI subgroup, 448 patients were randomized to ticagrelor-aspirin and 450 to placebo; among them, 187 (20.8%) had DWI-negative strokes (97 [21.7%] versus 90 [20.0%]), and 711 (79.2%) had single acute infarction (351 [78.3%] versus 360 [80.0%]). Within the MAI subgroup, 205 (50.1%) were allocated to ticagrelor-aspirin and 204 (49.9%) to placebo. Compared with patients without MAI, those with MAIs were older (median, 67.6 years [IQR, 59.7–73.4] versus 64.8 years [IQR, 58.0–71.0]), more often male (76.5% versus 68.8%), and had lower median BMI (23.0 kg/m² [IQR, 21.3–25.7] versus 24.2 kg/m² [IQR, 22.1–26.4]). Ischemic heart disease was more common in the MAI group (14.9% versus 9.8%), and baseline NIHSS score was slightly higher (median 6 [IQR, 6–8] versus 6 [IQR, 5–7]). Stroke cause also differed by infarction pattern: large artery atherosclerosis was more frequent in the MAI group (55.0% versus 35.7%), whereas small vessel occlusion predominated in the non-MAI group (60.2% versus 37.7%). Randomization occurred before thrombolysis in 197 (15.1%) patients, during in 169 (12.9%), and after in 941 (72.0%). Median onset-to-randomization time was 240 minutes in both groups, with no substantial differences in thrombolytic agent, onset-to-IVT time, IVT-to-randomization time, vascular risk factors, prior antiplatelet use, or baseline mRS score (Table 1). Within each infarction stratum, baseline characteristics were well balanced between treatment arms (Table 2).

Off-protocol antiplatelet agents were used, with tirofiban serving as the agent for protocol-defined rescue therapy despite no restrictions on agent selection. During days 0 to 7, tirofiban was administered to 15 of 205 (7.3%) of patients with MAI in the ticagrelor-aspirin group versus 37 of 204 (18.1%) in the placebo group, compared with 32 of 448 (7.1%) versus 38 of 450 (8.4%) among patients without MAI (Table S2). During days 8 to 90, off-protocol antiplatelet use was low and

Table 1. Baseline Characteristics of the Overall Population by Infarction Pattern

Characteristics	Non-MAIs (n=898)	MAIs (n=409)	P value
Age, y	64.8 (58.0–71.0)	67.6 (59.7–73.4)	0.0007*
Sex, n (%)			0.0043*
Female	280 (31.2)	96 (23.5)	
Male	618 (68.8)	313 (76.5)	
BMI, kg/m ²	24.2 (22.1–26.4)	23.0 (21.3–25.7)	0.0002*
Blood pressure, mm Hg			
Systolic	154.0 (140.0–167.0)	152.0 (139.0–165.0)	0.10
Diastolic	88.0 (80.0–96.0)	86.0 (78.0–95.0)	0.09
Current smoking	296 (33.0)	151 (36.9)	0.16
Current drinking	137 (15.3)	73 (17.8)	0.24
Antiplatelet use within 1 mo	87 (9.7)	33 (8.1)	0.35
Medical history			
Ischemic stroke	199 (22.2)	104 (25.4)	0.19
Ischemic heart disease	88 (9.8)	61 (14.9)	0.007*
Diabetes	209 (23.3)	95 (23.2)	0.99
Hypertension	528 (58.8)	259 (63.3)	0.12
Baseline mRS score			0.56
0	817 (91.0)	373 (91.2)	
1	78 (8.7)	36 (8.8)	
2	3 (0.3)	0 (0.0)	
Baseline NIHSS score	6.0 (5.0–7.0)	6.0 (6.0–8.0)	0.0035*
Thrombolytic agent			0.08
rt-PA	416 (46.3)	211 (51.6)	
TNK	482 (53.7)	198 (48.4)	
TOAST classification			<0.0001*
Large artery atherosclerosis	321 (35.7)	225 (55.0)	
Cardioembolic	4 (0.4)	6 (1.5)	
Small vessel occlusion	541 (60.2)	154 (37.7)	
Other determined cause	10 (1.1)	10 (2.4)	
Undetermined cause	22 (2.4)	14 (3.4)	
Onset-to-IVT time, min	154.0 (110.0–210.0)	160.0 (115.0–213.0)	0.46
IVT-to-randomization time, min	66.0 (17.0–112.0)	67.0 (20.0–113.0)	0.53
Onset-to-randomization time, min	240.0 (170.0–297.0)	240.0 (180.0–300.0)	0.20
Timing of randomization			0.68
Before intravenous thrombolysis	139 (15.5)	58 (14.2)	
During intravenous thrombolysis	112 (12.5)	57 (13.9)	
After intravenous thrombolysis	647 (72.0)	294 (71.9)	

Data are presented as median (interquartile range) for continuous variables and n (%) for categorical variables. BMI indicates body mass index; IVT, intravenous thrombolysis; MAIs, multiple acute infarctions; mRS, modified Rankin Scale; NIHSS, National Institutes of Health Stroke Scale; rt-PA, recombinant tissue plasminogen activator; TNK, tenecteplase; and TOAST, Trial of Org 10172 in Acute Stroke Treatment.

balanced across treatment groups and infarction patterns (Table S2).

Prognostic Impact of Infarction Patterns

Patients with MAIs had lower rates of excellent functional outcome at 90 days (241 [58.9%] of 409 versus 621 [69.2%] of 898; adjusted RR, 0.92 [95% CI,

0.84–1.00]; $P=0.05$) and functional independence (70.7% versus 81.2%; aRR 0.92 [95% CI, 0.86–0.98]; $P=0.01$) than those with non-MAIs. Ordinal analysis of the mRS score distribution also demonstrated a shift toward greater disability in the MAI group (adjusted common odds ratio, 0.72 [95% CI, 0.58–0.90]; $P=0.004$; Figure 2A). Early neurological improvement at 7 days was less frequent in patients with MAIs than

Table 2. Baseline Characteristics of Treatment Groups Stratified by Infarction Pattern

Characteristics	Non-MAIs	Placebo (n=450)	P value	MAIs	Placebo (n=204)	P value
	Early ticagrelor-aspirin (n=448)			Early ticagrelor-aspirin (n=205)		
Age, y	63.7 (57.6–70.5)	65.4 (58.4–71.8)	0.13	67.0 (58.7–73.4)	68.0 (59.9–73.5)	0.43
Sex, n (%)			0.36			0.15
Female	146 (32.6)	134 (29.8)		42 (20.5%)	54 (26.5%)	
Male	302 (67.4)	316 (70.2)		163 (79.5)	150 (73.5)	
BMI, kg/m ²	24.0 (22.0–26.1)	24.3 (22.2–26.8)	0.11	23.0 (21.2–25.8)	23.4 (21.4–25.7)	0.83
Blood pressure, mm Hg	154.0 (140.0–167.5)	153.0 (140.0–167.0)	0.90	152.0 (139.0–165.0)	152.5 (140.0–164.0)	0.90
Systolic	88.0 (79.0–96.0)	88.0 (80.0–96.0)	0.80	85.0 (78.0–96.0)	86.0 (78.0–95.0)	0.75
Diastolic	24.0 (22.0–26.1)	24.3 (22.2–26.8)	0.11	23.0 (21.2–25.8)	23.4 (21.4–25.7)	0.83
Current smoking	145 (32.4)	151 (33.6)	0.70	77 (37.6)	74 (36.3)	0.79
Current drinking	77 (17.2)	60 (13.3)	0.11	39 (19.0)	34 (16.7)	0.53
Antiplatelet use within 1 mo	48 (10.7)	39 (8.7)	0.30	18 (8.8)	15 (7.4)	0.60
Medical history						
Ischemic stroke	97 (21.7)	102 (22.7)	0.71	55 (26.8)	49 (24.0)	0.51
Ischemic heart disease	40 (8.9)	48 (10.7)	0.38	31 (15.1)	30 (14.7)	0.91
Diabetes	104 (23.2)	105 (23.3)	0.97	46 (22.4)	49 (24.0)	0.71
Hypertension	267 (59.6)	261 (58.0)	0.63	130 (63.4)	129 (63.2)	0.97
Baseline mRS score			0.87			0.60
0	406 (90.6)	411 (91.3)		185 (90.2)	188 (92.2)	
1	40 (8.9)	38 (8.4)		20 (9.8)	16 (7.8)	
2	2 (0.5)	1 (0.2)		0 (0.0)	0 (0.0)	
Baseline NIHSS score	6.0 (5.0–7.0)	6.0 (6.0–7.0)	0.42	6.0 (6.0–8.0)	6.0 (5.0–8.0)	0.57
Thrombolytic agent			0.16			0.96
rt-PA	218 (48.7)	198 (44.0)		106 (51.7)	105 (51.5)	
TNK	230 (51.3)	252 (56.0)		99 (48.3)	99 (48.5)	
TOAST classification			0.99			0.49
Large artery atherosclerosis	164 (36.6)	157 (34.9)		111 (54.1)	114 (55.9)	
Cardioembolic	2 (0.4)	2 (0.4)		3 (1.5)	3 (1.5)	
Small vessel occlusion	266 (59.4)	275 (61.1)		81 (39.5)	73 (35.8)	
Other determined cause	5 (1.1)	5 (1.1)		6 (2.9)	4 (2.0)	
Undetermined cause	11 (2.5)	11 (2.4)		4 (2.0)	10 (4.9)	
Onset-to-IVT time, min	154.0 (110.0–210.0)	152.0 (110.0–207.0)	0.64	159.0 (118.0–213.0)	160.0 (110.0–212.0)	0.85
IVT-to-randomization time, min	65.0 (10.0–110.0)	67.0 (22.0–114.0)	0.32	71.0 (31.0–117.0)	65.5 (9.0–110.0)	0.18
Onset-to-randomization time, min	240.0 (170.0–290.0)	240.0 (172.0–300.0)	0.52	250.0 (185.0–300.0)	240.0 (174.5–300.0)	0.27
Timing of randomization			0.22			0.42
Before intravenous thrombolysis	76 (17.0)	63 (14.0)		25 (12.2)	33 (16.2)	
During intravenous thrombolysis	61 (13.6)	51 (11.4)		27 (13.2)	30 (14.7)	
After intravenous thrombolysis	311 (69.4)	336 (74.6)		153 (74.6)	141 (69.1)	

Data are presented as median (interquartile range) for continuous variables and n (%) for categorical variables. BMI indicates body mass index; IVT, intravenous thrombolysis; MAIs, multiple acute infarctions; mRS, modified Rankin Scale; NIHSS, National Institutes of Health Stroke Scale; rt-PA, recombinant tissue plasminogen activator; TNK, tenecteplase; and TOAST, Trial of Org 10172 in Acute Stroke Treatment.

in those with non-MAIs (51.6% versus 58.5%; adjusted RR, 0.88 [95% CI, 0.79–0.98]; $P=0.02$). Severe adverse events (11.5% versus 3.7%; adjusted RR, 2.68 [95% CI, 1.74–4.12]; $P<0.001$) and any adverse events (20.0% versus 12.1%; adjusted RR, 1.56 [95% CI, 1.20–2.03]; $P<0.001$) were more common in the

MAI group. Recurrent ischemic stroke and sICH were numerically higher among patients with MAIs, but these associations were not statistically significant after adjustment, and the latter had wide CIs. No significant association was observed between infarction pattern and any bleeding within 90 days (Table 3).

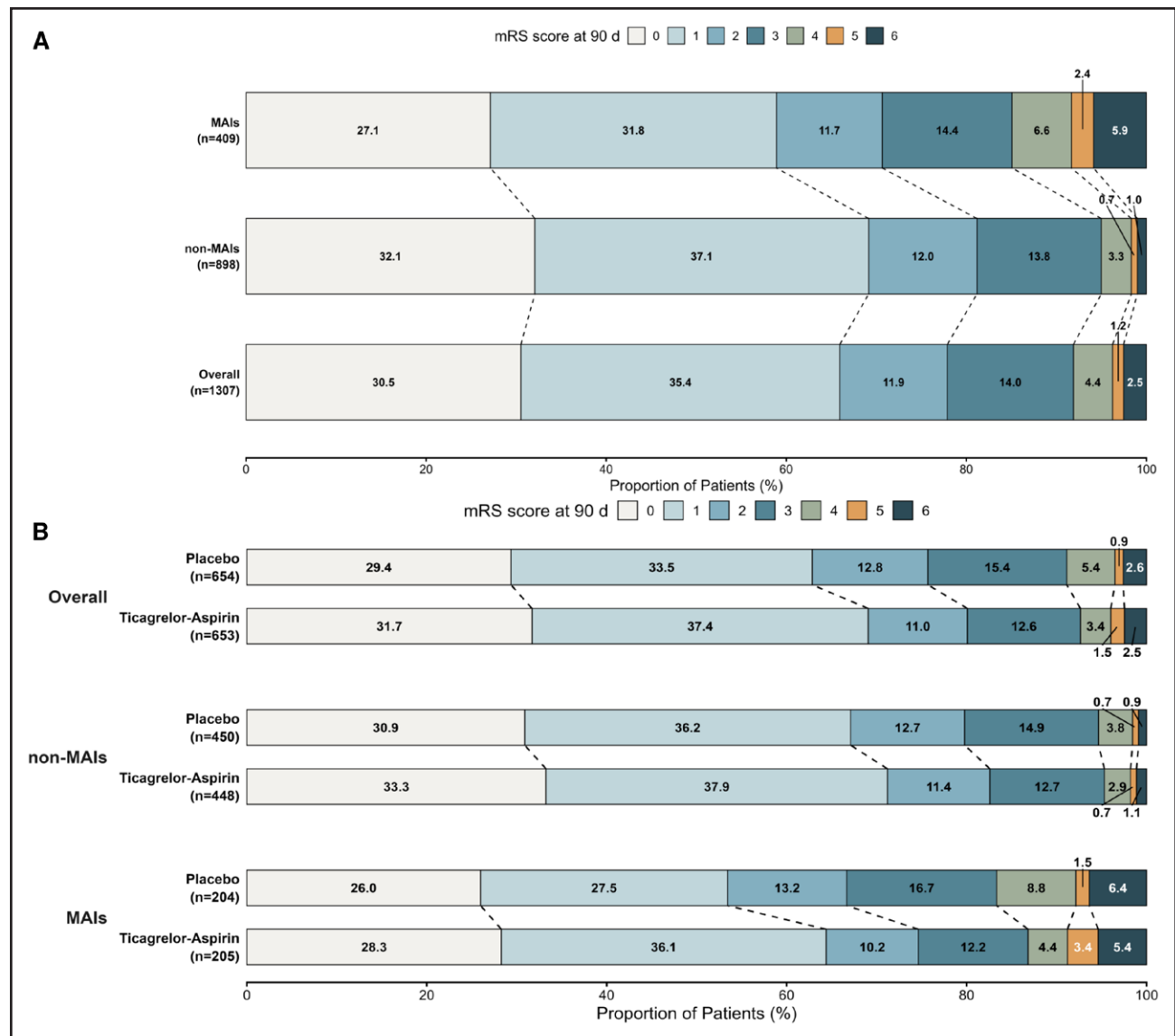


Figure 2. Distribution of modified Rankin Scale (mRS) scores at 90 days in the intention-to-treat population.

This conceptual illustration depicts diffusion-weighted imaging infarction patterns in acute ischemic stroke after intravenous thrombolysis. Multiple acute infarcts (MAIs) indicate higher thromboembolic vulnerability. Early ticagrelor and aspirin showed consistent benefits in this subgroup, although the primary interaction was not statistically significant. These hypothesis-generating findings support future studies.

Efficacy Outcomes

Among patients with MAIs, excellent functional outcome at 90 days (mRS score 0–1) occurred in 132 (64.4%) patients receiving early ticagrelor-aspirin and 109 (53.4%) receiving placebo (RR, 1.21 [95% CI, 1.02–1.42]; $P=0.0256$). In the non-MAI subgroup, corresponding rates were 320 (71.2%) and 301 (67.1%; RR 1.06 [95% CI, 0.97–1.16]; $P=0.18$; Figure 2B). The interaction between infarction pattern and treatment effect on the primary outcome was not statistically significant ($P_{\text{interaction}}=0.1788$). Among secondary efficacy outcomes, ticagrelor-aspirin increased the likelihood of early neurological improvement at 7 days in the MAI subgroup, with NIHSS score improvement of at least 4 points observed

in 118 (57.6%) patients in the ticagrelor-aspirin group and 93 (45.6%) in the placebo group (RR, 1.26 [95% CI, 1.04–1.53]; $P=0.02$). Significant heterogeneity according to infarction pattern was observed for this end point ($P_{\text{interaction}}=0.02$). In the non-MAI subgroup, early neurological improvement occurred in 259 (57.8%) patients receiving ticagrelor-aspirin and 265 (59.1%) receiving placebo (RR, 0.98 [95% CI, 0.88–1.09]; $P=0.69$). Directionally consistent but nonsignificant trends favoring ticagrelor-aspirin in the MAI subgroup were observed for functional independence at 90 days, ordinal mRS score distribution, and Barthel Index ≥ 95 at 90 days. EQ-5D index scores and EQ-5D VAS scores at 90 days were similar between treatment groups across infarction patterns. Rates of recurrent ischemic stroke within 90 days

Table 3. Clinical Outcomes Compared Between Patients With Non-MAIs and MAIs

Clinical Outcomes	Non-MAIs (n=898)	MAIs (n=409)	Model 1 effect size (95% CI)*	P value	Model 2 effect size (95% CI)†	P value
Primary outcome						
Good functional outcome (mRS score 0–1), RR	621 (69.2%)	241 (58.9%)	0.86 (0.78–0.94)	<0.001	0.92 (0.84–1.00)	0.05
Secondary outcomes						
Functional independence (mRS score 0–2), RR	729 (81.2%)	289 (70.7%)	0.87 (0.81–0.94)	<0.001	0.92 (0.86–0.98)	0.01
Ordinal mRS score at 90 d, cOR	...	...	0.64 (0.52–0.79)	<0.001	0.72 (0.58–0.90)	0.004
Early neurological improvement at 7 d, RR	525 (58.5%)	211 (51.6%)	0.89 (0.79–0.99)	0.03	0.88 (0.79–0.98)	0.02
Recurrent ischemic stroke within 90 d, HR	18 (2.0%)	26 (6.4%)	1.02 (0.91–1.15)	0.73	1.02 (0.90–1.15)	0.77
slCH at 36 h (ECASS III criteria)	2 (0.2%)	3 (0.7%)	3.30 (0.55–19.69)	0.19	2.75 (0.45–16.85)	0.27
slCH at 7 d (ECASS III criteria)	2 (0.2%)	5 (1.2%)	5.49 (1.07–28.24)	0.04	4.43 (0.83–23.62)	0.08
slCH at 36 h (SITS-MOST criteria)	2 (0.2%)	3 (0.7%)	3.30 (0.55–19.69)	0.19	2.75 (0.45–16.85)	0.27
slCH at 7 d (SITS-MOST criteria)	2 (0.2%)	4 (1.0%)	4.40 (0.81–23.96)	0.09	3.57 (0.64–19.88)	0.15
Any bleeding within 90 d (GUSTO criteria), RR	55 (6.1%)	26 (6.4%)	1.04 (0.66–1.63)	0.88	1.04 (0.66–1.64)	0.86
Severe adverse events within 90 d, RR	33 (3.7%)	47 (11.5%)	3.13 (2.03–4.80)	<0.001	2.68 (1.74–4.12)	<0.001
Any adverse events within 90 d, RR	109 (12.1%)	82 (20.0%)	1.65 (1.27–2.14)	<0.001	1.56 (1.20–2.03)	<0.001

Data are presented as n (%) unless otherwise indicated. All effect estimates are presented with 95% CIs. cOR indicates common odds ratio; ECASS III, European Cooperative Acute Stroke Study III; GUSTO, Global Utilization of Streptokinase and Tissue Plasminogen Activator for Occluded Coronary Arteries; HR, hazard ratio; MAI, Multiple Acute Infarction; mRS, modified Rankin Scale; NIHSS, National Institutes of Health Stroke Scale; RR, risk ratio; slCH, symptomatic intracranial hemorrhage; and SITS-MOST, Safe Implementation of Thrombolysis in Stroke-Monitoring Study.

*Model 1 adjusted for treatment group, age, sex, baseline NIHSS score, and TOAST classification.

†Model 2 additionally adjusted for history of ischemic stroke and BMI (baseline variables with $P \leq 0.10$). Firth penalized likelihood was used to account for sparse event data.

also did not differ significantly between treatment groups in either subgroup (Table 4).

Safety Outcomes

In the MAI subgroup, slCH according to ECASS III criteria within 36 hours occurred in 1 (0.5%) patient receiving ticagrelor-aspirin and 2 (1.0%) receiving placebo; corresponding rates at 7 days were 2 (1.0%) and 3 (1.5%). In the non-MAI subgroup, 1 patient (0.2%) in each treatment group experienced slCH within both 36 hours and 7 days (Table 4). Similar findings were observed using the SITS-MOST definition. No significant between-group differences were observed for any bleeding, adverse events, or severe adverse events within 90 days in either infarction-pattern subgroup.

Sensitivity and Exploratory Analyses

Per-protocol analyses were consistent with the primary findings (Table S3). In the MAI subgroup, ticagrelor-aspirin was associated with higher rates of excellent functional outcome, functional independence, early neurological improvement, and Barthel Index ≥ 95 at 90 days, whereas no significant treatment effect was observed in the non-MAI subgroup. Heterogeneity according to infarction pattern remained evident for early neurological improvement at 7 days ($P_{\text{interaction}} = 0.02$).

In a sensitivity analysis accounting for rescue antiplatelet therapy, a nominally significant treatment-by-infarction pattern interaction for the primary outcome

was observed ($P = 0.04$). Within the MAI subgroup, early ticagrelor-aspirin significantly improved the 90-day good functional outcome compared with placebo (62.9% versus 47.5%; adjusted RR, 1.32 [95% CI, 1.11–1.58]; $P = 0.002$), whereas no significant treatment effect was observed in the non-MAI subgroup (69.0% versus 64.2%; adjusted RR, 1.07 [95% CI, 0.98–1.18]; $P = 0.13$; Table S4). In exploratory analyses stratified by the timing of randomization relative to thrombolysis (Table S5), the treatment effect in the MAI subgroup appeared pronounced among patients randomized after thrombolysis (65.4% versus 48.9%; RR, 1.34 [95% CI, 1.09–1.64]; $P = 0.006$), with no statistically significant benefit observed in the before- or during-thrombolysis strata. No significant treatment effects were observed in any timing stratum among patients without MAI. The treatment-by-timing interaction was not significant in either the MAI ($P = 0.21$) or non-MAI ($P = 0.84$) subgroup. Exploratory analyses within the non-MAI subgroup showed no significant benefit of ticagrelor-aspirin across DWI-negative stroke or individual single-infarction phenotypes, including PAI, single small subcortical infarction, and branch atheromatous disease (all $P > 0.05$; Table S6).

DISCUSSION

Among patients with moderate non-cardioembolic ischemic stroke treated with IVT, those with MAIs had worse 90-day functional outcomes and a higher adverse event rates than those without MAIs. Early oral ticagrelor-aspirin was associated with a higher likelihood of

Table 4. Efficacy and Safety Outcomes of Early Antiplatelet Therapy by Infarction Pattern in the Intention-to-Treat Population

Outcome	Non-MAIs				MAIs				<i>P</i> _{interaction}
	Early ticagrelor-aspirin (n=448)	Placebo (n=450)	Effect size (95% CI)	<i>P</i> value	Early ticagrelor-aspirin (n=205)	Placebo (n=204)	Effect size (95% CI)	<i>P</i> value	
Primary efficacy outcome									
mRS score 0 to 1 at 90 d, RR	319 (71.2%)	302 (67.1%)	1.06 (0.97–1.16)	0.1848	132 (64.4%)	109 (53.4%)	1.21 (1.02–1.42)	0.0256	0.1788
Secondary efficacy outcomes									
mRS score 0–2 at 90 d, RR	370 (82.6%)	359 (79.8%)	1.04 (0.97–1.10)	0.28	153 (74.6%)	136 (66.7%)	1.12 (0.99–1.27)	0.08	0.28
Ordinal mRS score at 90 d, cOR	...	...	1.16 (0.91–1.47)	0.22	...	...	1.33 (0.94–1.89)	0.11	0.36
0	149 (33.3%)	139 (30.9%)			58 (28.3%)	53 (26.0%)			
1	170 (37.9%)	163 (36.2%)			74 (36.1%)	56 (27.5%)			
2	51 (11.4%)	57 (12.7%)			21 (10.2%)	27 (13.2%)			
3	57 (12.7%)	67 (14.9%)			25 (12.2%)	34 (16.7%)			
4	13 (2.9%)	17 (3.8%)			9 (4.4%)	18 (8.8%)			
5	3 (0.7%)	3 (0.7%)			7 (3.4%)	3 (1.5%)			
6	5 (1.1%)	4 (0.9%)			11 (5.4%)	13 (6.4%)			
NIHSS score decreasing ≥4 at 7 d, RR	259 (57.8%)	266 (59.1%)	0.98 (0.88–1.09)	0.69	118 (57.6%)	93 (45.6%)	1.26 (1.04–1.53)	0.02	0.02
Barthel Index ≥95 at 90 d, RR	371 (82.8%)	355 (78.9%)	1.05 (0.98–1.12)	0.15	155 (75.6%)	137 (67.2%)	1.13 (0.99–1.27)	0.06	0.31
EQ-5D score at 90 d, MD	0.9±0.2	0.9±0.3	0.02 (–0.01–0.05)	0.28	0.8±0.4	0.8±0.4	0.05 (–0.03–0.12)	0.25	0.44
EQ-5D VAS at 90 d, MD	83.4±18.5	81.7±19.3	1.68 (–0.80–4.15)	0.18	78.1±25.5	75.1±26.7	3.02 (–2.04–8.08)	0.25	0.64
Recurrent ischemic stroke within 90 d, HR	11 (2.5%)	7 (1.6%)	1.59 (0.62–4.10)	0.34	15 (7.3%)	11 (5.4%)	1.49 (0.67–3.33)	0.33	0.95
Safety outcomes									
Symptomatic ICH (ECASS III)									
At 36 h	1 (0.2%)	1 (0.2%)	NE	...	1 (0.5%)	2 (1.0%)	NE	...	...
At 7 d	1 (0.2%)	1 (0.2%)	NE	...	2 (1.0%)	3 (1.5%)	NE	...	...
Symptomatic ICH (SITS-MOST)									
At 36 h	1 (0.2%)	1 (0.2%)	NE	...	1 (0.5%)	2 (1.0%)	NE	...	...
At 7 d	1 (0.2%)	1 (0.2%)	NE	...	1 (0.5%)	3 (1.5%)	NE	...	...
Any bleeding within 90 d (GUSTO)	31 (6.9%)	24 (5.3%)	1.30 (0.77–2.18)	0.32	12 (5.9%)	14 (6.9%)	0.85 (0.39–1.86)	0.68	...
Minor	28 (6.3%)	21 (4.7%)			6 (2.9%)	10 (4.9%)			
Moderate	1 (0.2%)	0 (0.0%)			1 (0.5%)	0 (0.0%)			
Severe/life-threatening	2 (0.4%)	3 (0.7%)			5 (2.4%)	4 (2.0%)			
Any adverse events within 90 d	61 (13.6%)	48 (10.7%)	1.28 (0.89–1.82)	0.18	39 (19.0%)	43 (21.1%)	0.88 (0.54–1.43)	0.60	...
Severe adverse events within 90 d	18 (4.0%)	15 (3.3%)	1.21 (0.61–2.36)	0.59	24 (11.7%)	23 (11.3%)	1.04 (0.61–1.78)	0.89	...

*P*_{interaction} was obtained by adding a treatment-by-subgroup term to the generalized linear model. CIs for secondary outcomes are not adjusted for multiplicity; therefore, these data are considered descriptive only. cOR indicates common odds ratio; ECASS III, European Cooperative Acute Stroke Study III; EQ-5D, EuroQol 5-dimension questionnaire; GUSTO, Global Utilization of Streptokinase and Tissue Plasminogen Activator for Occluded Coronary Arteries; HR, hazard ratio; ICH, intracranial hemorrhage; MAIs, Multiple Acute Infarctions; MD, mean difference; mRS, modified Rankin Scale; NE, not estimable; NIHSS, National Institutes of Health Stroke Scale; RR, risk ratio; sICH, symptomatic intracranial hemorrhage; and SITS-MOST, Safe Implementation of Thrombolysis in Stroke-Monitoring Study.

excellent functional outcome and early neurological improvement in the MAI subgroup. However, the treatment-by-subgroup interaction for the primary outcome

was not statistically significant ($P=0.18$). In a post-hoc sensitivity analysis adjusting for rescue tirofiban use, the interaction was nominally significant ($P=0.04$). Neither

comparison was adjusted for multiplicity. Therefore, while the direction and consistency of treatment effects across secondary end points suggest a potential benefit of early oral ticagrelor-aspirin in patients with MAIs, these findings are exploratory and should be interpreted as hypothesis-generating.

This analysis extends the primary findings of the TAPIS trial, which showed that early oral ticagrelor-aspirin initiated around the time of IVT was associated with improved functional outcomes in the overall population without an apparent increase in sICH. In the present imaging substudy, the absolute risk difference for excellent functional outcome was 11.0% in the MAI subgroup (number needed to treat = 9), compared with a benefit of only 4.1% in the non-MAI subgroup. The direction of treatment effect was consistent across non-MAIs subtypes; the smaller magnitude may reflect a ceiling effect, as these patients already had favorable outcomes with placebo (67.1% achieved mRS score 0–1). Previous exploratory analyses from landmark antiplatelet trials have reported imaging-based heterogeneity for dual antiplatelet therapy in nonthrombolized patients with minor stroke, with benefit concentrated among those with MAIs or DWI-positive lesions.^{13,14,18} The present analysis extends these observations to the thrombolysis setting, where the balance between antithrombotic efficacy and hemorrhagic risk is particularly critical.^{20,23} Safety outcomes were comparable across treatment groups, with low sICH rates observed in this thrombolized cohort of patients with moderate stroke and without high baseline bleeding risk.

The worse prognosis observed in MAIs patients is likely rooted in the distinct pathophysiological mechanisms underlying different infarction patterns. In this study, large artery atherosclerosis was the predominant stroke cause in the MAIs subgroup (55.0%), whereas small vessel occlusion predominated in patients without MAIs (60.2%). Multiple acute lesions on DWI typically reflect active embolic sources, friable clot fragmentation, or highly unstable proximal plaques. These pathological states are associated with intense local platelet activation, a high thrombus burden, and an elevated risk of continuous microembolization or early reocclusion.^{10,13,24} Ticagrelor provides rapid and potent P2Y₁₂ inhibition, and aspirin blocks complementary thromboxane-mediated pathways.²⁵ Early oral administration of dual antiplatelet therapy may therefore stabilize the culprit lesion, suppress ongoing thromboembolism during the vulnerable early phase after reperfusion. In Asia, ~60% of patients carry CYP2C19 loss-of-function alleles,^{26,27} which may enhance the relative benefit of ticagrelor compared with clopidogrel. Conversely, patients without MAIs primarily present with single, localized lesions arising from intrinsic small vessel disease. These small subcortical lesions are more commonly associated with lipohyalinosis, which carries a lower inherent risk of early propagation or acute

recurrence. Consequently, functional outcomes in this subgroup may be determined by the severity of the initial parenchymal injury rather than by subsequent platelet-mediated events.^{28–30}

In the MAI subgroup, rescue tirofiban was administered more frequently to placebo-treated patients than to those receiving ticagrelor-aspirin (18.1% versus 7.3% during days 0–7). This imbalance suggests that patients with MAI on placebo were more susceptible to early neurological deterioration, and may have diluted the observed between-group difference in the primary analysis by improving outcomes in the placebo group through rescue intervention. In the analysis adjusting for rescue tirofiban use, the treatment-by-infarction pattern interaction for the primary outcome was nominally significant. Although unadjusted for multiplicity, this finding, together with the significant interaction for early neurological improvement at 7 days ($P=0.02$), is consistent with the biological plausibility that early ticagrelor-aspirin may suppress ongoing microembolization in patients with MAI. These findings are insufficient to change current practice, but may provide a rationale for future enrichment trials to prospectively test DWI infarction pattern as a biomarker for treatment selection.

The non-MAI group comprised DWI-negative stroke and single acute infarction as distinct phenotypes. This heterogeneity should be considered when interpreting the comparison with the MAI group. The binary MAI versus non-MAI classification was chosen to prioritize clinical feasibility in the hyperacute setting, supported by high inter-rater agreement ($\kappa=0.88$). DWI-negative strokes, which accounted for 14.3% of the overall population, were included for operational reasons; isolating them would have introduced statistical instability and diluted power for other infarction subtypes. Single acute infarctions constituted the majority of the non-MAI group (79.2%), enabling further subclassification by lesion morphology as prespecified. Although pooling diverse phenotypes may have diluted treatment effects, exploratory analyses across individual non-MAI subtypes did not show a clear treatment signal. This heterogeneity also points to future directions; targeted trials could evaluate high-risk variants such as branch atheromatous disease separately from lower-risk mechanisms like lipohyalinosis-driven single small subcortical infarctions.

Limitations

Several limitations should be acknowledged. First, infarction patterns were classified using conventional DWI criteria without quantitative assessment of lesion volume, clot burden, or perfusion status. Second, the non-MAI group is biologically heterogeneous, which may have diluted treatment effects, as discussed above. Third,

although between-center variability in magnetic resonance imaging acquisition timing was observed, adjusted analyses did not demonstrate a significant association between imaging timing and DWI lesion detectability; nevertheless, delayed imaging in some centers may have influenced detection of smaller lesions. Fourth, the study population was exclusively Chinese, where intracranial atherosclerosis prevalence differs substantially from Western populations, which limits generalizability. Fifth, the parent trial excluded patients with severe stroke, planned endovascular thrombectomy, atrial fibrillation, or anticoagulant use; the findings may therefore not be generalizable to those subsets. Finally, systematic vascular imaging follow-up was not performed, so direct evidence that the observed treatment benefit resulted from reduced reocclusion or decreased thrombus burden is lacking.

Conclusions

In patients with moderate non-cardioembolic ischemic stroke (NIHSS score 4–10) treated with IVT within 4.5 hours of onset, those with MAIs had poorer 90-day functional outcomes. Among these patients, early oral ticagrelor-aspirin initiated within 6 hours of onset (before, during, or after thrombolysis) was associated with directionally consistent improvements across efficacy outcomes, without an apparent increase in bleeding risk. However, the treatment-by-subgroup interaction for the primary outcome was not statistically significant. In a sensitivity analysis adjusting for rescue tirofiban use, the interaction was nominally significant, further supporting the potential benefit in this subgroup, though not adjusted for multiplicity. Given the exploratory nature of these analyses, the findings are hypothesis-generating and may inform future enrichment trials evaluating diffusion-weighted imaging infarction pattern as an imaging biomarker to guide antiplatelet therapy in the early reperfusion setting.

ARTICLE INFORMATION

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Supplemental Material

eAppendix
Tables S1–S6
Figure S1
Appendices S1–S6
Trial Protocol and Statistical Analysis Plan
CONSORT Checklist



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